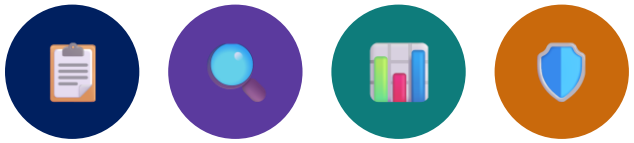
MODULE 20

Logging Frameworks and Diagnostics

Explore enterprise-grade logging frameworks, structured logging, and diagnostics techniques for reliable production systems.



TEKsystems Global Services

WORKFORCE DEVELOPMENT

MODULE 20 OVERVIEW

Logs are one major observability signal. They complement metrics and traces by providing detailed event context.

- This module explores enterprise-grade logging frameworks, structured logging, and diagnostics techniques to improve observability, troubleshoot issues, and ensure reliable production systems.

DURATION & LAB

Theory

SLF4J, Logback, log levels, structured logging.

Lab

Structured Logging in a Spring Boot app.

Scenario

Add production-ready structured logs with correlation IDs.

BUILD

SLF4J + Logback structured logging

OBSERVE & DIAGNOSE

Correlation IDs, exceptions, centralized logs

THE PAYOFF

Faster troubleshooting, reliable systems

KEY TAKEAWAY You can't improve what you can't see — logs turn data into actionable insights.

WHY LOGGING MATTERS

Logs are the primary window into what your application is actually doing in production.

Visibility

See what's really happening inside a running application.

Faster Diagnosis

Pinpoint root causes without reproducing the issue live.

Monitoring

Feed dashboards and alerts that catch problems early.

Auditability

Provide a durable record for security and compliance reviews.

Request Correlation

Connect related log events across services using correlation or trace context.

Faster Recovery

Shorten mean time to resolution during incidents.

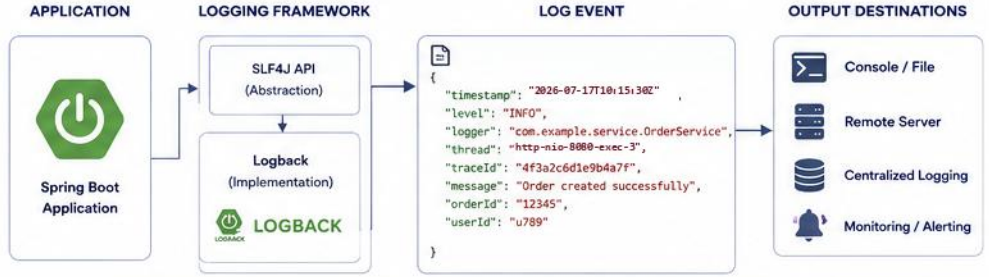
Good logs. Better insights. Reliable systems. Log data also supports security/compliance audits and informed operational decisions. Real distributed tracing (trace ID, span ID, parent-child spans, service graph) is covered in Module 21.

KEY TAKEAWAY You can't improve what you can't see — logs turn data into actionable insights.

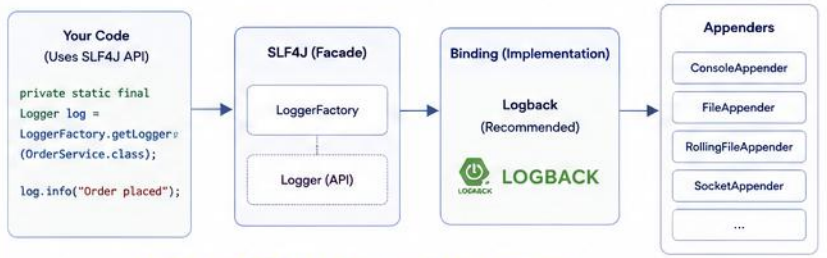
LOGGING IN MODERN APPLICATIONS

Capture the right information at the right level to understand, troubleshoot and improve your system.

1. LOGGING FLOW / ARCHITECTURE



2. LOGGING FRAMEWORK STACK



You write once (SLF4J), run with any compatible implementation.

3. LOG LEVELS EXPLAINED

Level	Use When	Severity
TRACE	Detailed trace information	100
DEBUG	Detailed debug information	200
INFO	General operational events	300
WARN	Something unexpected happened but application continues	400
ERROR	A serious problem occurred	500
OFF	Disables logging entirely	—

Least Severe → Most Severe

4. STRUCTURED LOGGING EXAMPLE (JSON)

5. CORRELATION & REQUEST FLOW

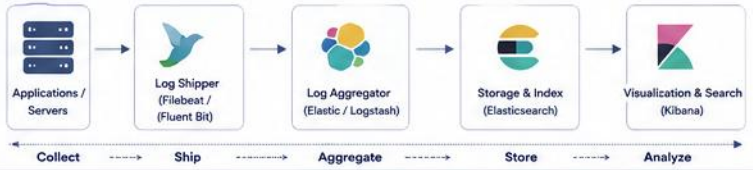
Makes it easy to trace a request across services and logs.

6. LOGGING EXCEPTIONS

```
try {
   orderService.processOrder(req);
} catch (PaymentFailedException ex) {
   log.error("Payment failed for orderId={}",
             req.getOrderId(), ex);
}
```

- Log with appropriate level (ERROR)
- Include exception stack trace
- Include context (IDs, input, state)
- Avoid logging sensitive data

7. CENTRALIZED LOGGING CONCEPT



8. PRODUCTION DIAGNOSTICS

- Troubleshoot issues faster
- Analyze trends & patterns
- Detect errors & anomalies
- Audit & compliance
- Understand user behavior

9. LOGGING BEST PRACTICES

Log the right things
Log meaningful events, not everything.

Use appropriate log levels
Avoid overusing ERROR or DEBUG in production.

Be consistent
Use consistent format, fields, and naming.

Include context
Add IDs (correlationId, userId, orderId, etc.)

Avoid sensitive data
Never log passwords, tokens, PII.

Structured logging
Use JSON for better search & analytics.

Review & rotate logs
Set retention, rotation and archive policies.

10. BENEFITS

- Faster troubleshooting
- Better monitoring & alerting
- Improved system reliability
- Operational visibility
- Audit & regulatory compliance

LOGGING VS DEBUGGING

Debugging and logging solve different problems — one is for development, the other for production.

DEBUGGING	LOGGING
• Interactive, step-by-step inspection	• Passive, always-on record of events
• Requires attaching a debugger to the process	• No special tooling needed to capture
• Only practical in development environments	• Works everywhere, including production
• Great for deep-diving a single reproducible issue	• Great for understanding behavior over time and across requests

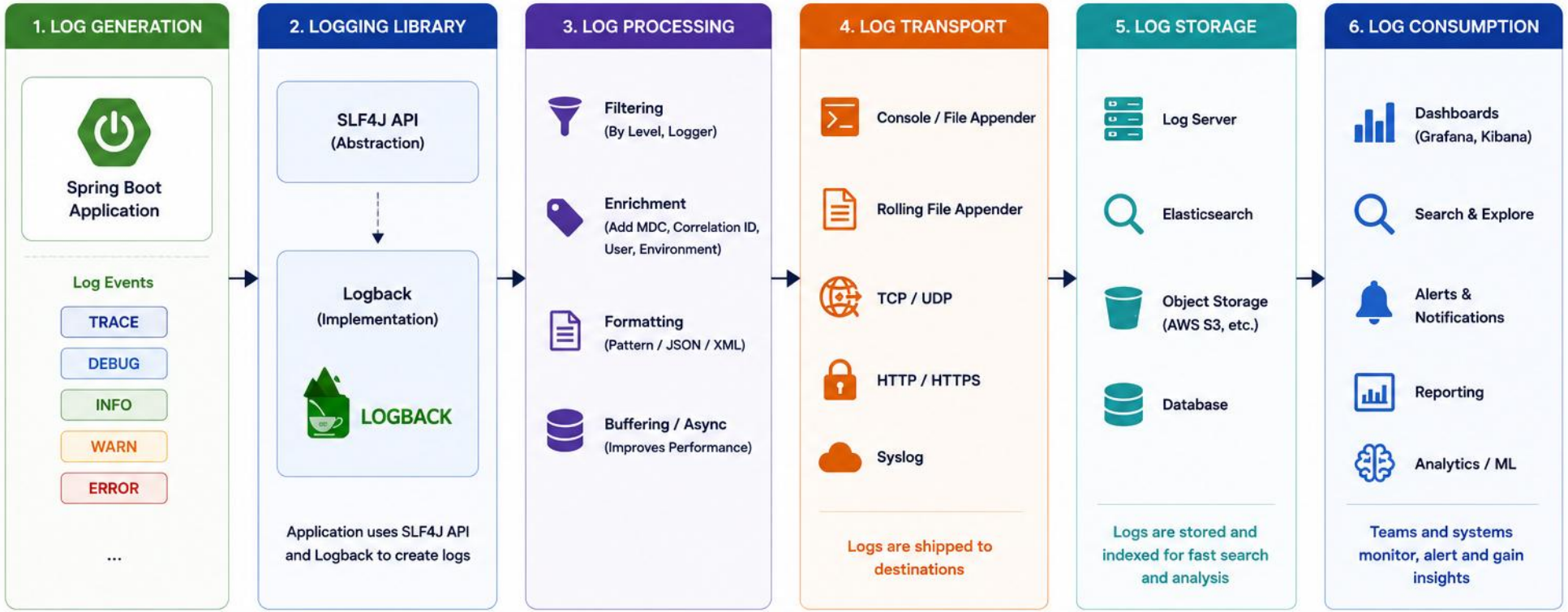
VS

Interactive debugging is generally unsuitable for normal production troubleshooting because it can pause execution, alter timing, and create security risk. Logs and production-safe diagnostics (diagnostic snapshots, tightly-controlled remote debugging, Java Flight Recorder, heap/thread dumps, profilers, debug containers) are the default and the exception, respectively. Logging can be production-safe when levels, volume, formatting, and transport are designed carefully.

KEY TAKEAWAY Debugging pauses execution and doesn't scale to production — logs and production-safe diagnostics are the default tools there.

LOGGING ARCHITECTURE

High-level view of how logs are generated, processed, transported, stored and consumed.



ENTERPRISE LOGGING PIPELINE

Application → SLF4J → Logback/Appender → Collector → Storage/Index → Search/Dashboard/Alert — one pipeline; actual tooling varies by organization.

STAGE	EXAMPLE TOOLS	PURPOSE
Application	Application / service code	Generate log events
SLF4J	Logging facade API	Format and handle events
Logback / Appender	Logging backend + appender	Send events to destinations
Collector	Agent / collector	Collect logs from many sources
Storage / Index	Searchable storage / archive storage	Index logs for search & retention
Search / Dashboard / Alert	Dashboard / alert destination	Turn logs into dashboards & alerts

LAYER	TYPICAL OWNER
Log event design	Application team
Application logging configuration	Application/platform
Collection and transport	Platform/SRE
Storage and retention	Platform/security
Dashboards and alerts	SRE/application teams

- Categories are illustrative — actual tooling varies by org. Example stack: SLF4J → Logback → Fluent Bit → Kafka → OpenSearch → Kibana.
- Async appenders cut request-thread blocking for high-volume systems but add queueing, a dropping policy, shutdown flushing, memory use, and ordering considerations.

KEY TAKEAWAY Real logging pipelines chain multiple specialized tools — producers to dashboards — into one observability platform.

INTRODUCTION TO SLF4J

SLF4J (Simple Logging Facade for Java) is an abstraction layer over concrete logging frameworks.

EXAMPLE

```
private static final Logger log =
    LoggerFactory.getLogger(OrderService.class);

log.info("Order {} placed for user {}",
    orderId, userId);

if (log.isDebugEnabled()) {
    log.debug("Customer snapshot {}", expensiveSnapshot());
}
```

WHY A FACADE?

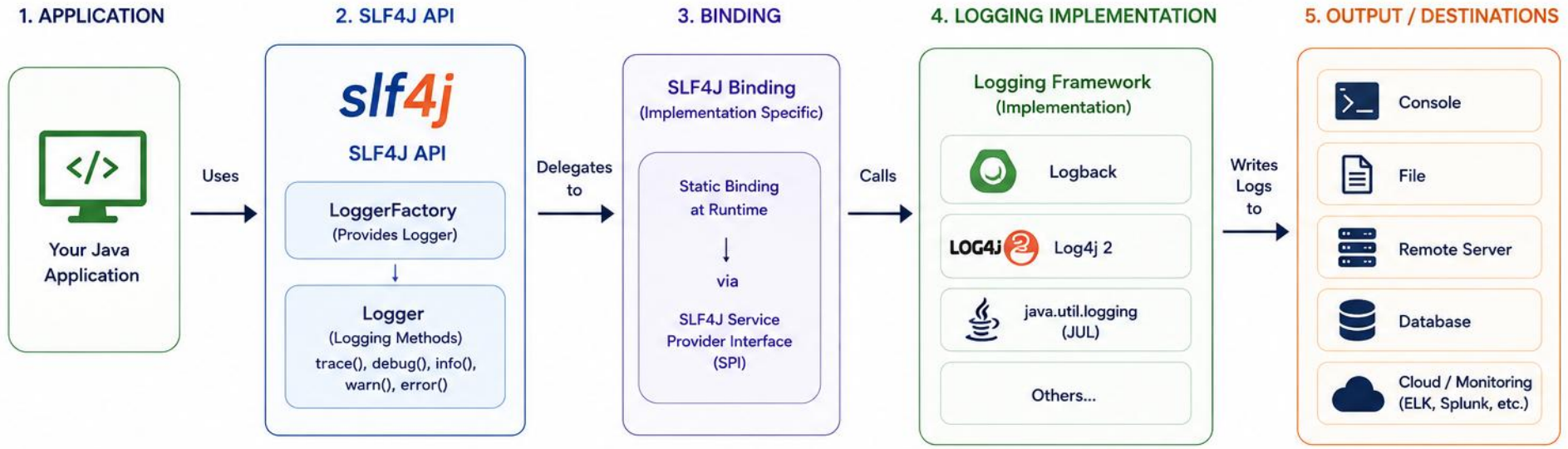
- Decouples your code from a specific logging implementation.
- SLF4J keeps application code independent of the backend API, making implementation changes easier without rewriting logging calls.
- Provides efficient parameterized logging (no string concatenation).
- The de facto standard logging API for Java.
- Keep one active SLF4J provider on the runtime classpath — multiple providers or logging bridges can cause warnings, duplicate events, or recursion.

"Write once, log anywhere." Common bindings: Logback (default), Log4j 2, java.util.logging, and others. Placeholders reduce unnecessary string construction; guard genuinely expensive arguments with a level check like isDebugEnabled() — use sparingly, not for every call.

KEY TAKEAWAY Code against the SLF4J API, not a specific logging framework — it keeps your logging backend swappable.

SLF4J OVERVIEW

SLF4J (Simple Logging Facade for Java) is a façade or abstraction for various logging frameworks.
It allows applications to remain independent of the underlying logging implementation.



HOW SLF4J WORKS

Example Dependency (Maven)

slf4j-api (Compile Time) + slf4j-logback-classic (Runtime Binding) = Logging Works with Logback

KEY POINTS

- SLF4J is just an API (facade), not a logging framework.
- It decouples the application from the logging implementation.
- You can switch the underlying logging framework without changing application code.
- Provides a simple and uniform logging abstraction.

INTRODUCTION TO LOGBACK

Logback is the most common SLF4J implementation, and the default in Spring Boot.

EXAMPLE: logback-spring.xml

```
<configuration>
    <appender name="CONSOLE" class="ch.qos.logback.core.ConsoleAppender">
        <encoder>
            <pattern>%d{ISO8601} %-5level [%thread] %logger{36} corr=%X{correlationId:-none} - %msg%n</pattern>
        </encoder>
    </appender>
    <root level="INFO">
        <appender-ref ref="CONSOLE"/>
    </root>
</configuration>
```

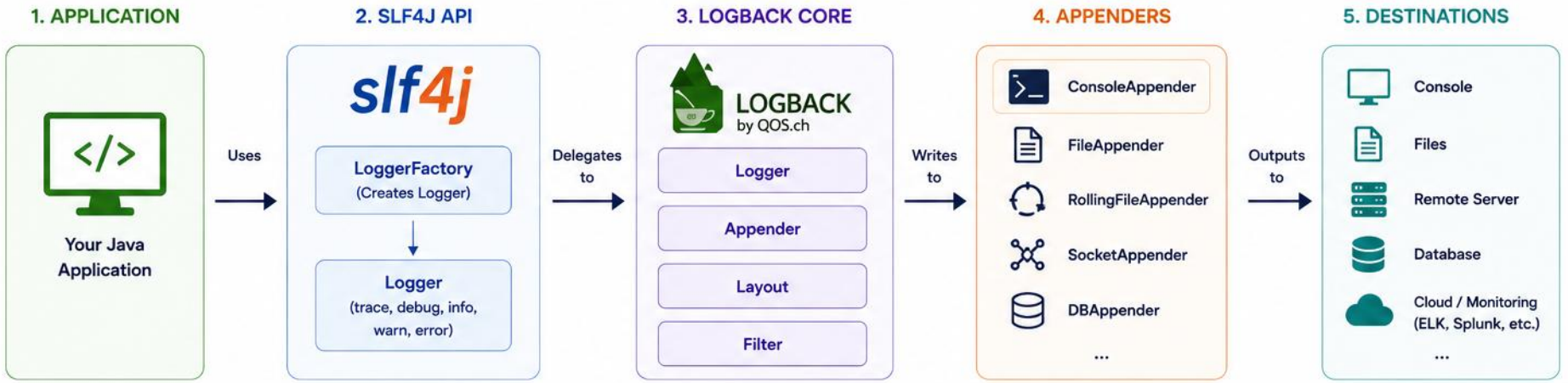
Logback is a mature SLF4J backend and the default logging implementation in many Spring Boot versions. Log4j 2 is another modern alternative. Console or file output collected asynchronously by the platform is common — direct network, database, or email appenders require careful reliability and performance design.

KEY TAKEAWAY logback.xml (or logback-spring.xml) configures appenders, patterns, and log levels for your whole application.

LOGBACK OVERVIEW

Logback is a powerful, high-performance logging framework for Java.
It is a successor of Log4j and a native implementation of SLF4J.

LOGBACK ARCHITECTURE



LOGBACK CONFIGURATION

XML

logback.xml

```
<configuration>
  <appender name="FILE" class="ch.qos.logback.core.rolling.RollingFileAppender">
    <file>logs/app.log</file>
    <rollingPolicy class="ch.qos.logback.core.rolling.SizeAndTimeBasedRollingPolicy">
      <fileNamePattern>logs/app.%d{yyyy-MM-dd}.%i.log</fileNamePattern>
      <maxFileSize>10MB</maxFileSize>
      <maxHistory>30</maxHistory>
    </rollingPolicy>
    <encoder>
      <pattern>%d{yyyy-MM-dd HH:mm:ss.SSS} %-5level [%thread] %logger - %msg%n</pattern>
    </encoder>
  </appender>

  <root level="INFO">
    <appender-ref ref="FILE" />
  </root>
</configuration>
```

LOGGING FLOW IN LOGBACK

KEY COMPONENTS OF LOGBACK

Logger
Entry point for logging. Organized in a hierarchy.

Appender
Responsible for publishing logs to various destinations.

Layout
Formats the log message before it is written by appender.

Filter
Decides whether a log event should be processed or ignored.

Configuration
Defined in logback.xml. Controls loggers, appenders, filters, and others.

LOG LEVELS EXPLAINED

Log levels let you control verbosity and filter logs by severity.

LEVEL	MEANING	WHEN TO USE
ERROR	Unexpected failure requiring investigation, or a failure causing the request/system to fail	Exceptions, failed operations
WARN	Abnormal condition handled successfully but worth monitoring	Deprecated usage, fallback behavior
INFO	Meaningful lifecycle or business event useful in normal operations	Startup, request received, order placed
DEBUG	Diagnostic detail for troubleshooting	Local development, deep troubleshooting
TRACE	Highly granular flow detail	Rarely enabled, very deep debugging

- Do not log every 4xx as an application error — use metrics for volume, security audit logs where appropriate, and reserve stack traces for unexpected defects.
- Use INFO for low-volume, operationally significant events; high-volume request details may belong in access logs, structured telemetry, sampled logs, or DEBUG depending on policy.

KEY TAKEAWAY Use the right level for the right audience — INFO for operations, DEBUG for developers, ERROR for incidents.

LOG LEVELS EXPLAINED

Log levels define the severity and importance of log events.
They help control what gets logged and what gets ignored.

SEVERITY	LOG LEVEL	DESCRIPTION	TYPICAL USE
MOST SEVERE	ERROR	Very severe events that cause application failures.	Exceptions, system crashes, data loss, critical failures
	WARN	Potentially harmful situations that don't stop the application but need attention.	Deprecated API usage, configuration issues, resource limits
	INFO	Informational messages that highlight the progress of the application.	Application startup/shutdown, user logins, business events
	DEBUG	Detailed diagnostic information for developers.	Variable values, method calls, application flow
	TRACE	Very detailed trace of the application flow.	Function entry/exit, step-by-step tracing
LEAST SEVERE	OFF	Disables all logging.	Used to turn off logging completely

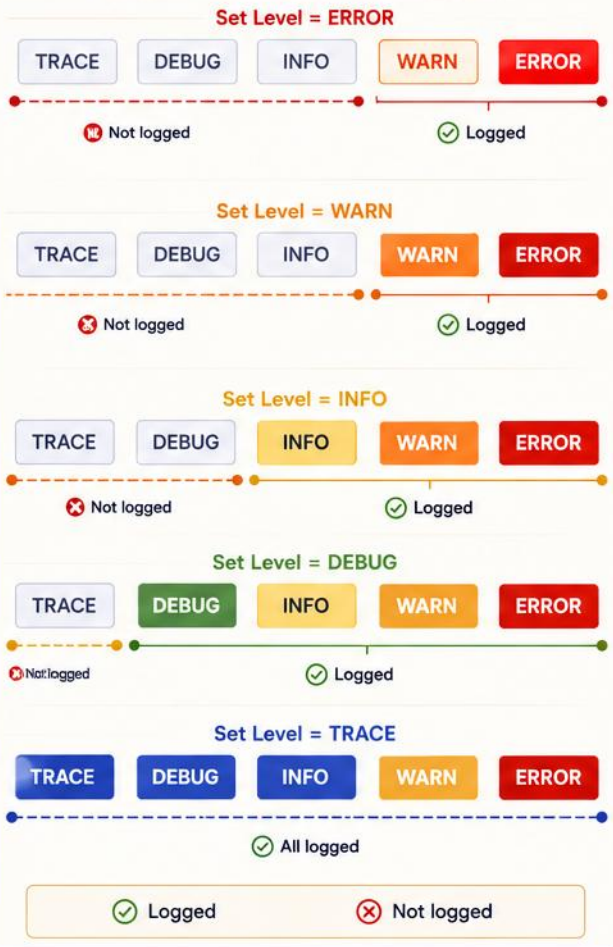
Important

Use the appropriate log level:

- High severity → ERROR, WARN
- Normal operational events → INFO
- Development & troubleshooting → DEBUG, TRACE
- Avoid using ERROR/WARN for normal flow

HOW LOG LEVEL FILTERING WORKS

The configured log level acts as a threshold.
All logs at that level and above are logged.



STRUCTURED LOGGING

Structured logging records events as named fields. JSON is a common transport format because collectors and search platforms can parse it reliably.

EXAMPLE (STRUCTURED EVENT)

```
{
  "timestamp": "2025-05-11T10:15:30.520Z",
  "level": "INFO",
  "logger": "OrderService",
  "event": "order.created",
  "operation": "create_order",
  "message": "Order placed",
  "correlationId": "3f9c2a6e7b4dd4e1a",
  "orderId": 10245,
  "outcome": "success"
}
```

AVOID DUPLICATING FIELDS IN THE MESSAGE

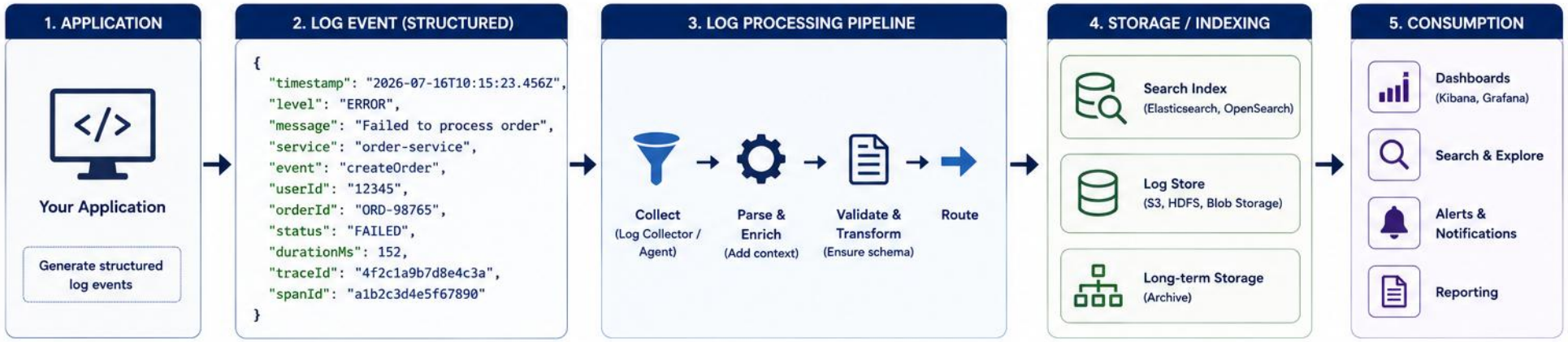
```
// A structured event should not rely on parsing the message:
log.atInfo()
   .addKeyValue("event", "order.created")
   .addKeyValue("orderId", orderId)
   .addKeyValue("outcome", "success")
   .log("Order created");
// Otherwise: use MDC or a structured encoder
```

- Use safe identifiers: an internal non-sensitive subject id where approved, a pseudonymous id, or customer ID only when allowed — never raw email, phone, name, session token, or credential. The lab's zero-PII objective should drive this theory.
- Checklist: stable event name, outcome, reason code, duration, service/environment, correlation or trace context. Common uses: microservices tracing, monitoring & alerting, audit/compliance trails.
- Event naming example: {"event": "customer.created", "operation": "create_customer", "customerId": "CUS-1001", "outcome": "success"}

KEY TAKEAWAY Free-form text becomes difficult to query consistently at scale — structured fields improve filtering, aggregation, and automated analysis.

STRUCTURED LOGGING

Logging events in a consistent, machine-readable format (key-value pairs) that is easy to search, filter, and analyze.



UNSTRUCTURED vs STRUCTURED LOG EXAMPLE

UNSTRUCTURED LOG (PLAIN TEXT)

```
2026-07-16 10:15:23 ERROR Failed to process order for user 12345
OrderService.createOrder(OrderService.java:45)
java.lang.NullPointerException: Order is null
```

- Hard to parse
- Inconsistent
- Difficult to search
- Harder to analyze

VS

STRUCTURED LOG (JSON)

```
{
"timestamp": "2026-07-16T10:15:23.456Z",
"level": "ERROR",
"message": "Failed to process order",
"service": "order-service",
"userId": "12345",
"orderId": "ORD-98765",
"status": "FAILED",
"traceId": "4f2c1a9b7d8e4c3a"
}
```

- Consistent format
- Easy to search
- Machine readable
- Easy to analyze

COMMON STRUCTURED LOG FIELDS

timestamp	level	message	service	userId	traceId	env
When the event occurred	Severity of the event	Short description of the event	Name of the application/service	User or client identifier	For distributed tracing	Environment (dev, test, prod)

COMMON STRUCTURED LOG FORMATS

JSON — Most popular, flexible and machine readable

XML — Extensible markup language

YAML — Human friendly, easy configuration

CORRELATION IDS

A correlation ID lets you trace a single request across every service and log entry it touches.

1 **Request Arrives**

Gateway generates a correlation ID.

2 **ID Propagated**

Passed via header to every downstream call.

3 **Included in Every Log**

Stored in MDC, added to each log entry.

4 **Trace End-to-End**

Search logs across services by ID.

EXAMPLE (MDC — VALIDATED & PROPAGATED)

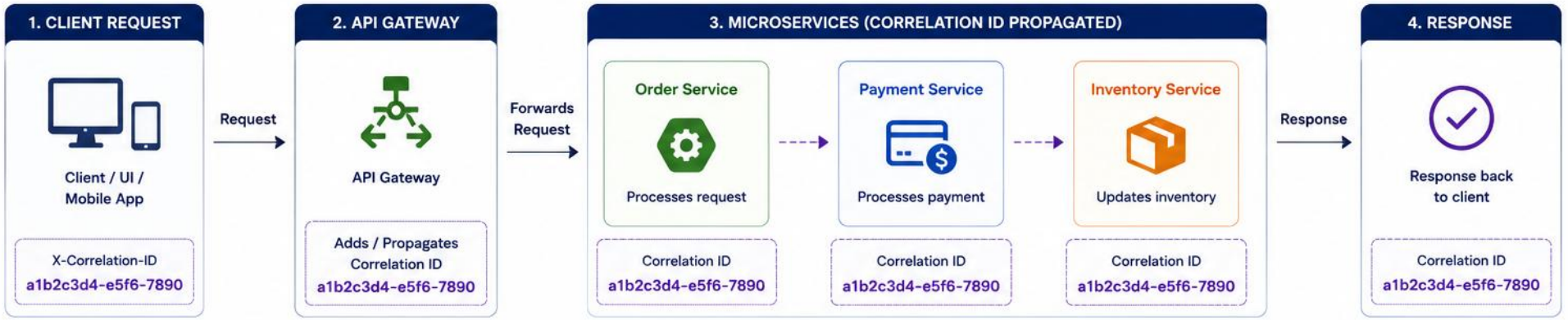
```
String incoming = request.getHeader("X-Correlation-Id");
String correlationId = isValid(incoming) ? incoming : newId();
try {
    MDC.put("correlationId", correlationId);
    response.setHeader("X-Correlation-Id", correlationId);
    log.info("Processing order request");
    downstream.propagate(correlationId);
} finally {
    MDC.remove("correlationId");
}
```

Use an opaque, validated identifier format defined by the organization — not just UUIDs (W3C trace IDs, ULIDs, and other opaque IDs are also used). Do not encode personal or security-sensitive data in it.
Ensure context propagation is supported when work moves across threads or reactive boundaries (thread pools, CompletableFuture, reactive pipelines, virtual threads). Always clear or restore MDC to prevent context leakage.
Correlation ID is application-level; trace ID belongs to distributed tracing (Module 21) — they may coexist, but a correlation ID in logs is not distributed tracing.

KEY TAKEAWAY Correlation IDs connect the dots across distributed systems — without them, tracing a request is guesswork.

CORRELATION IDs EXAMPLE

Correlation ID is a unique ID assigned to a request and included in logs across all services.
It helps trace the full request flow end-to-end.



The same Correlation ID (a1b2c3d4-e5f6-7890) is included in every log entry, allowing you to trace the entire request across multiple services and components.

5. LOGS WITH CORRELATION ID

Timestamp	Service	Level	Message	Correlation ID
2026-07-16 10:15:23.456	API Gateway	INFO	Request received: POST /api/orders	a1b2c3d4-e5f6-7890
2026-07-16 10:15:23.457	Order Service	INFO	Order creation started. orderId=ORD-98765	a1b2c3d4-e5f6-7890
2026-07-16 10:15:23.789	Payment Service	INFO	Payment initiated. amount=129.99	a1b2c3d4-e5f6-7890
2026-07-16 10:15:24.212	Payment Service	INFO	Payment successful. txnId=TXN-555123	a1b2c3d4-e5f6-7890
2026-07-16 10:15:24.518	Inventory Service	INFO	Inventory updated. orderId=ORD-98765	a1b2c3d4-e5f6-7890
2026-07-16 10:15:24.900	Order Service	INFO	Order created successfully. orderId=ORD-98765	a1b2c3d4-e5f6-7890
2026-07-16 10:15:25.001	API Gateway	INFO	Response sent: 201 Created	a1b2c3d4-e5f6-7890

HOW IT HELPS

- Track a request end-to-end across all services
- Easily identify where a failure occurred
- Measure latency at each step
- Improve debugging and monitoring

Always pass the Correlation ID in headers and include it in logs.

LOGGING REQUEST FLOWS

Log incoming requests and outgoing responses with enough context to reconstruct what happened.

WHAT TO INCLUDE

- HTTP method, normalized route (e.g. /customers/{id}), and status code — remove or allowlist query parameters.
- Response time / latency.
- Correlation ID for every entry.
- Never log sensitive fields — mask passwords and tokens.

EXAMPLE

```
log.info("{} {} -> {} ({} ms)",
    method, uri, status, elapsedMs);
// Business decision (structured):
{"event":"customer.activation",
 "outcome":"rejected",
 "reasonCode":"INVALID_STATUS_TRANSITION",
 "customerId":"CUS-1001"}
```

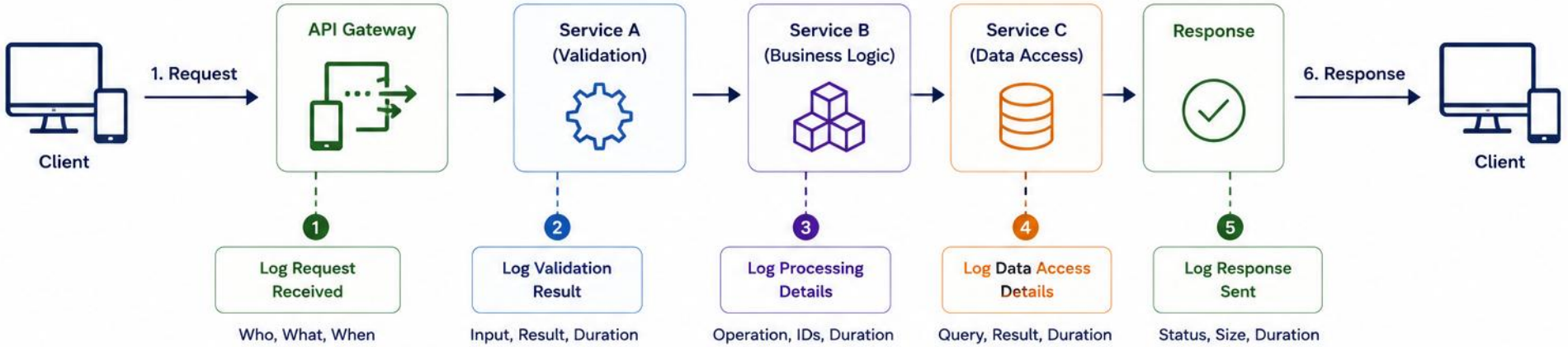
Also: use database-driver/framework diagnostics at DEBUG when troubleshooting; use metrics for latency and error rate; log business-level persistence outcomes; never log bind values containing secrets or PII; avoid N+1 troubleshooting through permanent high-volume logs.

Access logs (transport-level request summary, often already recorded by web servers/gateways/service meshes) differ from application/business logs (business events and decisions), audit logs (security/compliance actions), and trace spans (distributed timing/causality). Avoid duplicating at the application level what access logs already capture.

KEY TAKEAWAY Consistent request/response logging turns every API call into a searchable, traceable record.

LOGGING REQUEST FLOWS

Log at key points of a request's lifecycle to understand how it moves through the system.



EXAMPLE LOG ENTRIES

	Timestamp	Correlation ID	Service / Component	Log Level	Message	Key Details
1	2026-07-16 10:15:23.456	a1b2c3d4-e5f6-7890	API Gateway	INFO	Request received	method=POST, path=/api/orders, client=10.0.0.5
2	2026-07-16 10:15:23.462	a1b2c3d4-e5f6-7890	Service A	INFO	Validation successful	userId=12345, payloadSize=1.2KB, durationMs=6
3	2026-07-16 10:15:23.470	a1b2c3d4-e5f6-7890	Service B	INFO	Order created	orderId=ORD-98765, amount=129.99, durationMs=18
4	2026-07-16 10:15:23.487	a1b2c3d4-e5f6-7890	Service C	INFO	Database update successful	table=orders, rows=1, durationMs=11
5	2026-07-16 10:15:23.505	a1b2c3d4-e5f6-7890	API Gateway	INFO	Response sent	status=201, responseSize=512B, durationMs=49

KEY LOGGING POINTS IN A REQUEST FLOW

Request Received
Capture who made the request and what was requested.

Validation
Log validation outcome and important inputs.

Processing
Log business actions, IDs, and processing duration.

Data Access
Log database calls, queries, and results.

Response Sent
Log the response status, size, and total time taken.

Include Correlation ID in every log to trace the flow end-to-end across all services.

Common Fields:
timestamp, correlationId, service, level, message, duration, status

LOGGING EXCEPTIONS

Log exceptions with the full stack trace and enough context to find the root cause quickly.

EXAMPLE

```
// Lower layer: add typed context, do NOT log here
catch (OrderException ex) {
    throw new OrderProcessingException(orderId, ex);
}
// Boundary: log exactly once
@ExceptionHandler(OrderProcessingException.class)
void handle(OrderProcessingException ex) {
    log.error("Order failed id={}", orderId, ex);
}
```

BEST PRACTICES

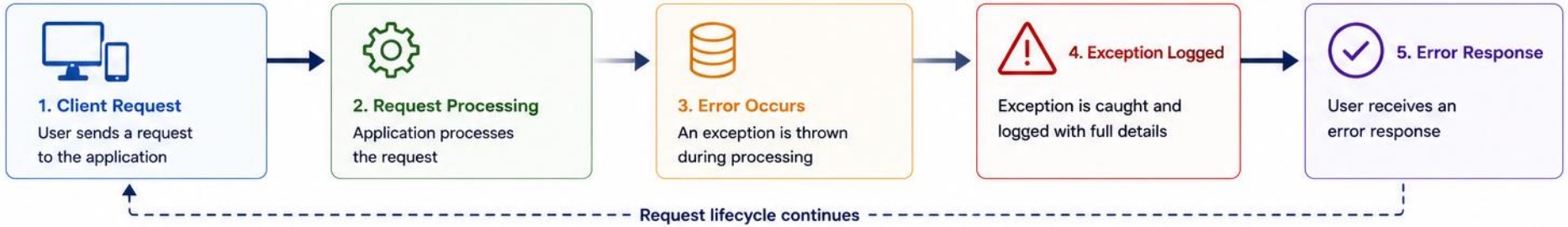
- Include the exception object when stack-trace diagnostics are useful and permitted. Avoid stack-trace noise for expected failures.
- Include the correlation ID and a safe identifier for the affected resource, following data-classification rules — don't expose sensitive information.
- Use ERROR level only for failures that need investigation or cause request/system failure.
- Log an exception at the boundary where sufficient context exists and where it will not be logged again. Lower layers should usually add typed context and rethrow without duplicate logging.

EXCEPTION CATEGORY	TYPICAL TREATMENT
Expected validation/domain rejection	Safe response, metric; usually no ERROR stack trace
Security event	Controlled audit record
Recoverable dependency issue	WARN or ERROR based on impact
Unexpected defect	ERROR with stack trace
Duplicate log from upstream boundary	Avoid

KEY TAKEAWAY A stack trace without context is half a clue — pair it with the correlation ID and relevant business data.

LOGGING EXCEPTIONS

Log exceptions with meaningful details to quickly identify, diagnose, and resolve issues.



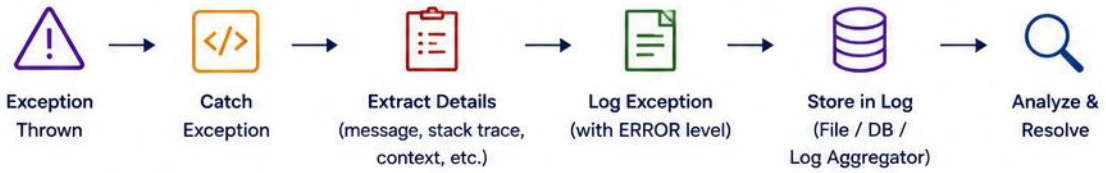
EXAMPLE: EXCEPTION LOG ENTRY

```
{
  "timestamp": "2026-07-16T10:15:23.456Z",
  "level": "ERROR",
  "thread": "http-nio-8080-exec-7",
  "logger": "com.example.order.service.OrderService",
  "message": "Failed to create order",
  "exception": "java.sql.SQLIntegrityConstraintViolationException",
  "exceptionMessage": "Duplicate entry 'ORD-98765' for key 'orders.PRIMARY'",
  "stackTrace": "java.sql.SQLIntegrityConstraintViolationException: Duplicate entry
    'ORD-98765' for key 'orders.PRIMARY'\n\tat
    com.mysql.cj.jdbc.exceptions.SQLError.createSQLException(SQLError.java:129)\n\tat
    com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:122)\n\tat
    com.mysql.cj.jdbc.ClientPreparedStatement.executeInternal(ClientPreparedStatement.java:953)\n\t... 23 more",
  "correlationId": "a1b2c3d4-e5f6-7890",
  "userId": "12345",
  "requestId": "req-8f7e6d5c",
  "service": "order-service",
  "environment": "production"
}
```

WHAT SHOULD BE LOGGED

When	Timestamp of the exception
What	Clear error message
Exception Type	Fully qualified exception class
Stack Trace	Full stack trace for root cause
Context	User ID, Request ID, Correlation ID, etc.
Where	Service/Component where error occurred
Environment	Environment (dev, test, prod, etc.)

EXCEPTION LOGGING FLOW



WHY IT MATTERS

- Faster root cause analysis
- Easier monitoring and alerting
- Better reliability and system stability
- Useful for audit and compliance

CENTRALIZED OPERATIONS AND GOVERNANCE

Collecting, storing, and visualizing logs from all services in one platform provides a major operational signal that complements metrics and traces.

Collection

Ship logs from every service to a central pipeline.

Storage & Indexing

Index log data so it stays fast to query at scale.

Visualization

Turn raw log data into dashboards teams can read.

Search Across Services

Find every entry for a request across all services at once.

Alerting

Notify the team automatically when error patterns spike.

Trend Analysis

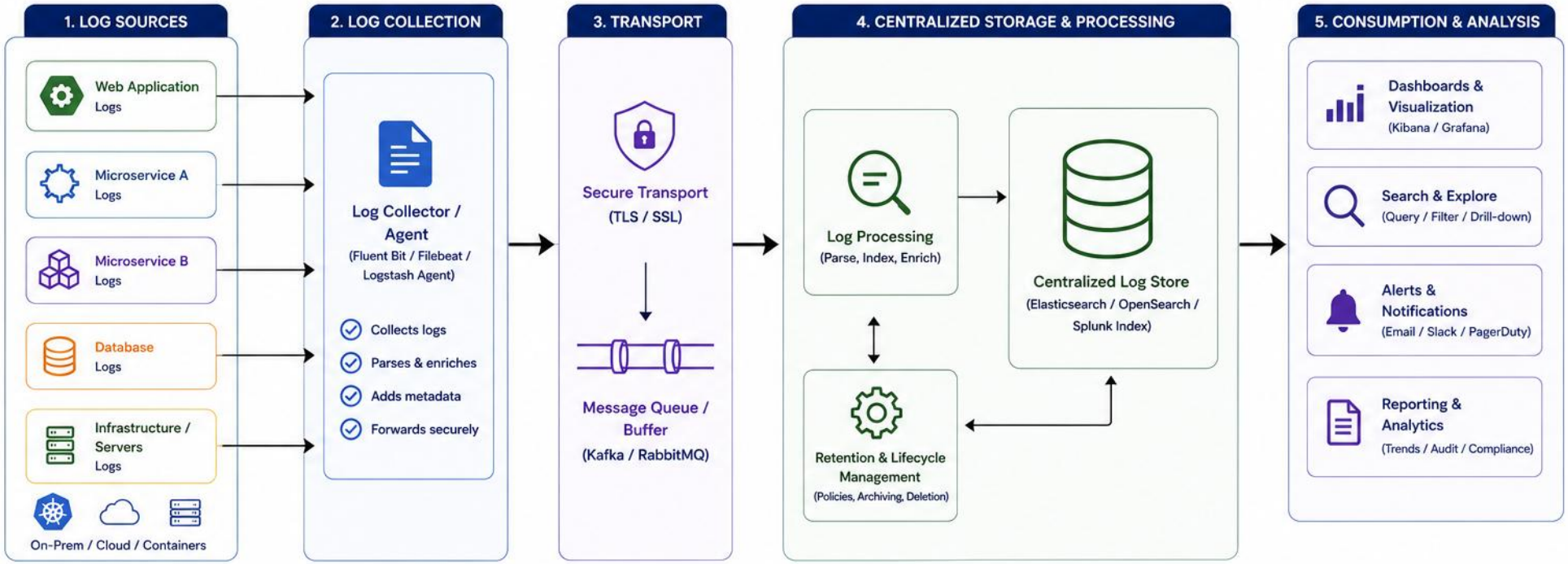
Spot recurring issues and usage patterns over time.

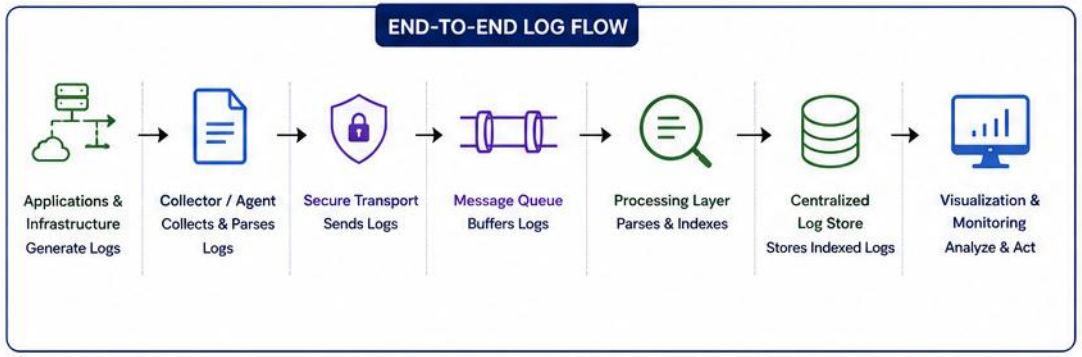
Typical stack: Fluent Bit (agent) → Logstash (pipeline) → Elasticsearch (index) → Kibana (visualize) → Alertmanager/Slack (notify) — illustrative; tooling varies by organization. Alert on actionable patterns and sustained conditions rather than every ERROR event. Better alert sources: error-rate metrics, aggregated log patterns, SLO burn rates, dependency failure rates, security rules.
Retention & governance: hot searchable retention, archive retention, deletion requirements, legal hold, cost controls, access control, and regional data residency.

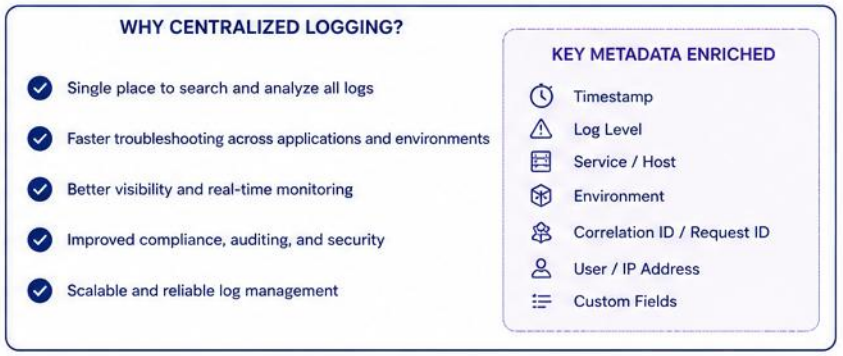
KEY TAKEAWAY Centralized logging provides observability at scale — searching one server's log files doesn't work in production.

CENTRALIZED LOGGING ARCHITECTURE

Collect logs from all applications and services in one place for unified storage, search, monitoring, and analysis.

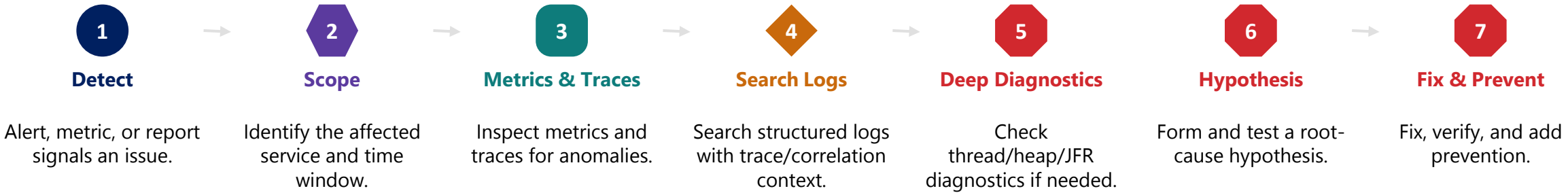






PRODUCTION DIAGNOSTICS

Logs are your primary tool for troubleshooting, root cause analysis, and understanding system health in production.



KEY PRACTICES

- Design logs around known operational questions while preserving enough safe context for investigation. Do not collect data merely because it might become useful.

Also track availability, performance, resource, error, and business metrics alongside logs — and always feed lessons learned back in.

KEY TAKEAWAY Design logs around known operational questions, not just to react to outages — diagnostics starts long before an incident.

PRODUCTION DIAGNOSTICS

Using logs for fast issue detection and root cause analysis in production environments.

1. ISSUE DETECTED

- User reports error
- Alert from monitoring
- Performance degradation

2. SEARCH LOGS

- Use Correlation ID / Request ID
- Filter by time, level, service, user
- Find related log entries

3. ANALYZE & CORRELATE

- Trace request flow across services
- Review structured logs & stack traces
- Correlate with metrics

4. IDENTIFY ROOT CAUSE

- Pinpoint failing component
- Understand error pattern
- Verify impact and scope

5. RESOLVE & PREVENT

- Apply fix / rollback
- Monitor logs for recovery
- Add alerts / logging improvements

EXAMPLE: DIAGNOSING A PRODUCTION ISSUE

Problem

Users report intermittent errors while placing orders.

Correlated Logs

Filter logs using:

Correlation ID:

`a1b2c3d4-e5f6-7890`

Relevant Log Entries

```
10:15:23.456  INFO   [order-service]
a1b2c3d4-e5f6-7890   Order request received

10:15:23.789  INFO   [payment-service]
a1b2c3d4-e5f6-7890   Calling payment API

10:15:24.512  ERROR  [payment-service]
a1b2c3d4-e5f6-7890   java.net.ConnectException:
                     Connection timed out

10:15:24.520  WARN   [order-service]
a1b2c3d4-e5f6-7890   Payment failed, retrying...
```

Root Cause

Payment service is having network connectivity issues (timeout).

Resolution

- Fix network / service issue
- Monitor for recovery
- Add alert for timeouts

KEY BENEFITS

- Faster root cause analysis
- Reduced mean time to resolution (MTTR)
- Better visibility into production issues
- Proactive problem detection
- Improved system reliability

PRODUCTION DIAGNOSTICS BEST PRACTICES

Use Correlation IDs
to trace requests end-to-end

Log structured data
for easy search and analysis

Include key context
(user, service, environment, duration)

Combine logs with metrics
and alerts for full visibility

Centralize and retain logs
for quick troubleshooting

LOGGING PRACTICES QUICK REFERENCE

A compact technical reference for five recurring practices — not a repeat of earlier explanations.

PRACTICE	DONE WHEN	EXAMPLE
`Log with Purpose`	Event is meaningful, not routine noise	Avoid logging every method entry/exit
`Consistent Format`	Same fields/structure across services	Shared schema (JSON or logfmt)
`Include Context`	Timestamp, level, correlation ID, safe identifiers present	Entry is self-explanatory without extra lookups
`Never Log Secrets`	Zero passwords, tokens, or raw PII in any field	Mask/allowlist fields — see hierarchy below
`Use the Right Level`	Level reflects operational meaning, not habit	Environment-specific defaults, controlled overrides

- Also: keep logs concise, watch the performance overhead of logging, plan retention/archival policies, and monitor your logging pipeline itself.
- Sensitive-data hierarchy: (1) do not collect, (2) allowlist safe fields, (3) tokenize/pseudonymize where needed, (4) redact as defense in depth — redaction can fail; partial masking may still expose regulated identifiers.
- Define environment-specific defaults and allow controlled, temporary diagnostic overrides for selected loggers.

KEY TAKEAWAY **These five practices turn logging from routine noise into a trustworthy operational signal.**

LAB OVERVIEW – STRUCTURED LOGGING

Add SLF4J + Logback structured logging with correlation IDs to the Northstar CRM customer service — with zero PII in logs.

WHAT YOU WILL BUILD

- A CorrelationFilter that validates/generates the ID, sets MDC + response header, propagates it downstream, and clears it in finally.
- Minimum: consistent key-value (logfmt) fields event=/operation=/outcome=/reasonCode=/customerId=/correlationId=/durationMs=. Preferred: JSON encoder with the same named fields.
- Instrumented CustomerService create/get calls — customer IDs only, never PII.

TECH STACK

ITEM	VALUE
`Framework`	Spring Boot
`Logging`	SLF4J + Logback
`Format`	Pattern layout (JSON optional)
`Tracing`	MDC correlation ID

QUOTE

- "Well-structured logs today become actionable insights tomorrow." Support can search what happened to a request without ever seeing a name, email, or phone number.

Scenario: extend the Customer Management Platform (Lab 19) so support can trace CUS-1001 / CUS-1002 by correlation ID.

KEY TAKEAWAY This lab builds the PII-free, correlation-traceable logging foundation centralized logging platforms depend on.

LAB TASKS AND DELIVERABLES

Add structured, PII-free logging with correlation IDs to the Northstar CRM customer service.

#	TASK	KEY ACTIVITIES	FOCUS
1	Branch Lab 19 & Confirm Logging	Copy to lab20-crm; confirm SLF4J + Logback binding.	Setup
2	Configure Structured Pattern	Build logback-spring.xml with corr/cust/op fields.	Structured logs
3	Add Correlation Filter	CorrelationFilter puts ID in MDC; clears in finally.	Tracing
4	Instrument Customer Service	Log create/get with customerId + op — never PII.	Request flows
5	Log Validation & Exceptions	Category-specific levels — not blanket WARN/ERROR (see verification notes).	Diagnostics
6	Verify with Automated Test	CustomerLoggingIT — ListAppender-based; asserts semantically; no PII.	Verification

DELIVERABLES

- lab20-crm project — structured, PII-free logging.
- logback-spring.xml — logfmt minimum, JSON preferred.
- CorrelationFilter — validate/generate, MDC, header, clear in finally.
- Instrumented CustomerService/Controller logging (no PII).
- CustomerLoggingIT — ListAppender-based, semantic no-PII proof.
- docs/logging.md contract; submit via course portal by due date.

SUCCESS CRITERIA & VERIFICATION

- Logs show corr=/cust=/op= for CUS-1001 and CUS-1002; CustomerLoggingIT passes; zero PII (names, emails, phones) in any log line; MDC cleared after every request.
- No-PII checks are strong evidence, not proof — layer fixture leak tests + code review.
- MDC-leak test: req A (corr-A) → req B (corr-B) → background log → confirm no corr-A leak.
- One exception, one stack trace — don't double-log; levels are category-specific, not blanket.
- Async propagation (if used): optional advanced exercise — verify cleanup.

KEY TAKEAWAY You will have PII-free, correlation-traceable logs that let support search "what happened to request X" with confidence.

PRODUCTION LOGGING DEFINITION OF DONE

A checklist for shipping logging that is safe, useful, and consistent in production.

1. Application uses SLF4J rather than backend-specific APIs.
2. Exactly one logging provider is configured.
3. Events have stable names and structured fields.
4. Levels reflect operational meaning.
5. Expected client errors do not generate noisy stack traces.
6. Unexpected failures are logged once.
7. Correlation context is validated, propagated, returned, and cleared.
8. Raw URLs, bodies, tokens, names, emails, and phone numbers are not logged.
9. Safe identifiers and reason codes are used.
10. Request access logs and business logs have distinct purposes.
11. Logs are centrally collected.
12. Retention and access policies are defined.
13. Logging volume and pipeline health are monitored.
14. Tests verify important events and privacy constraints.

KEY TAKEAWAY Meeting every item on this checklist is what makes logging production-ready — not just functional.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of logging frameworks and diagnostics.

1. What is SLF4J?

A. A logging backend implementation

B. A logging facade/abstraction over logging frameworks

C. A centralized log storage system

D. A Spring Boot annotation

2. Which log level is for unexpected but recoverable issues?

A. ERROR

B. WARN

C. INFO

D. TRACE

3. An API logs the full request URL, including ?email=user@example.com. What should change?

A. Nothing — full URLs are fine for debugging

B. Log the normalized route, not personal query values

C. Encrypt the entire URL before logging

D. Stop logging that endpoint entirely

4. Why prefer structured (JSON) logs?

A. They are shorter to write

B. They are easy to parse, search, and analyze

C. They remove the need for log levels

D. They only work in development

KEY TAKEAWAY SLF4J + Logback, the right log level, and correlation IDs are the foundation of production-ready logging.

KNOWLEDGE CHECK (2 OF 2)

Four more questions, plus the full answer key.

5. What should NEVER appear in a log entry?

A. A timestamp

B. A correlation ID

C. Passwords or tokens in plain text

D. A log level

6. Service and handler both log one exception — what's the main problem?

A. There is no problem — more logs are more thorough

B. Each layer should always keep its own audit trail

C. Duplicate noise and alerts — log it once at the boundary

D. It has no real downside

7. What should accompany an exception logged at ERROR?

A. Nothing else is needed

B. Exception object, correlation/trace context, safe details

C. Only the exception class name

D. A password reset link

8. What is the main goal of production diagnostics?

A. To avoid writing any logs

B. To use logs for troubleshooting and root cause analysis

C. To disable logging in production

D. To replace correlation IDs with debuggers

ANSWER KEY

- **1-B 2-B 3-B 4-B 5-C 6-C 7-B 8-B**

KEY TAKEAWAY Great logs lead to great insights, faster fixes, and stronger systems.

"You can't improve what you can't see. Logs turn data into actionable insights."

MODULE 20 COMPLETE

Logging Frameworks and Diagnostics

You can now build a practical, production-ready logging strategy using industry-standard tools and practices.